

graph#

<https://pytorch.org/docs/stable/generated/torch.cuda.graph.html#torch.cuda.g>

Description:

Context-manager that captures CUDA work into a `torch.cuda.CUDAGraph` object for later replay. See [CUDA Graphs](#) for a general introduction, detailed use, and constraints.

`cuda_graph` (`torch.cuda.CUDAGraph`) – Graph object used for capture.

`pool` (optional) – Opaque token (returned by a call to `graph_pool_handle()` or `other_Graph_instance.pool()`) hinting this graph's capture may share memory from the specified pool. See [Graph memory management](#).

`stream` (`torch.cuda.Stream`, optional) – If supplied, will be set as the current stream in the context. If not supplied, graph sets its own internal side stream as the current stream in the context.

Function Signature

```
class torch.cuda.graph(cuda_graph, pool=None, stream=None,
capture_error_mode='global')[source]#
```

Parameters

Notes & Warnings

NOTE

Note For effective memory sharing, if you pass a pool used by a previous capture and the previous capture used an explicit stream argument, you should pass the same stream argument to this capture.

⚠ WARNING

Warning This API is in beta and may change in future releases.

Detailed Documentation

Documentation

Context-manager that captures CUDA work into a `torch.cuda.CUDAGraph` object for later replay.

See [CUDA Graphs](#) for a general introduction, detailed use, and constraints.

`cuda_graph` (`torch.cuda.CUDAGraph`) – Graph object used for capture.

`pool` (optional) – Opaque token (returned by a call to `graph_pool_handle()` or `other_Graph_instance.pool()`) hinting this graph's capture may share memory from the specified pool. See [Graph memory management](#).

`stream` (`torch.cuda.Stream`, optional) – If supplied, will be set as the current stream in the context. If not supplied, graph sets its own internal side stream as the current stream in the context.

`capture_error_mode` (str, optional) – specifies the `cudaStreamCaptureMode` for the graph capture stream. Can be “global”, “thread_local” or “relaxed”. During cuda graph capture, some actions, such as `cudaMalloc`, may be unsafe. “global” will error on actions in other threads, “thread_local” will only error for actions in the current thread, and “relaxed” will not error on actions. Do NOT change this setting unless you’re familiar with `cudaStreamCaptureMode`

For effective memory sharing, if you pass a pool used by a previous capture and the previous capture used an explicit stream argument, you should pass the same stream argument to this capture.

This API is in beta and may change in future releases.